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QoE-Optimized DQN Admission Control for Network Slicing on a Live OAI O-RAN Testbed

Manoj Prasad Kunasegran, Wai Leong Pang*, and Swee King Phang

***Abstract*—Slice Admission Control (SAC) decides whether an incoming slice request can be accommodated given current resources. However, static SAC cannot adapt as user demands shift, particularly in dynamic 5G environments. Deep Reinforcement Learning (DRL)-driven SAC closes this gap by incorporating context-aware mechanisms such as Service Level Agreement (SLA) and Quality of Experience (QoE) into the admission policy. We evaluate a Deep Q-Network (DQN)-based approach on a real OpenAirInterface (OAI) gNodeB across three slices. These are Enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communication (URLLC) and Massive Machine-Type Communication (mMTC). The QoE-aware policy is benchmarked against a static baseline, an SLA-only reward, and a static-at-cap policy that pins every slice at its calibrated maximum allocation. After finalising QoE calibration, a fully powered live evaluation across 128 episodes per arm demonstrated the QoE-aware DQN sustaining 100.0% eMBB/URLLC/mMTC SLA compliance on every episode, a statistically significant improvement over the static baseline's 93.9% slice-wide compliance. The SLA-only reward and the static-at-cap policy converge to a statistically indistinguishable compliance rate, 98.6% and 98.7% slice-wide respectively, only numerically higher than the baseline. Five further SLA-reward checkpoints show that live robustness varies substantially by seed despite equivalent offline convergence. Under a congested, URLLC-preservation scenario, the DQN policy trades eMBB compliance for URLLC compliance offline. However, during live evaluation, every arm remains fully SLA-compliant. A smaller pilot still shows the QoE-aware reward giving URLLC a materially higher Mean Opinion Score of 4.83 out of 5, against 2.44 for the SLA-only reward and 2.66 for the static baseline. Negligible control-loop latency of 0.57 ms and a lightweight model under 20,000 parameters confirm minimal deployment overhead. These findings demonstrate the feasibility of deploying DRL-driven SAC in a live Open RAN architecture.**

Index Terms—5G, Deep Reinforcement Learning, Network Slicing, Open RAN, Quality of Experience, Slice Admission Control

I. Introduction and Background

Network slicing virtualises physical 5G deployments to serve heterogeneous classes under per-slice Service Level Agreements (SLAs). Ultra-Reliable Low-Latency Communication (URLLC) needs sub-millisecond latency, Enhanced Mobile Broadband (eMBB) needs high throughput, and Massive Machine-Type Communication (mMTC) serves many low-power devices. Slice Admission Control (SAC) decides whether an incoming request can be accommodated given current resources, and static SAC cannot adapt effectively as user demand changes over time.

Our prior work applied DRL to SAC under gNodeB (gNB) congestion, using SLA-driven reward shaping to prioritise URLLC traffic [1]. It later expanded to a third slice type, mMTC, with load balancing across multiple gNBs [2]. Both were validated only in simulation. Our systematic review [3] places this research space into three evidence tiers. These are literature-supported work, preliminary simulated studies and a proposed but not-yet-validated graph-attention multi-agent federated-learning (GAT-MARL-FL) framework. This identifies validated deployment as a gap. This paper closes that gap for the same formulation. We realise it on a real OpenAirInterface (OAI) gNB over its native E2 interface, the O-RAN near-real-time RAN Intelligent Controller (RIC) [4]. Only the radio channel remains simulated, a mode we denote rfsim throughout, and the reward is centred on QoE rather than SLA compliance alone. This paper makes the following contributions.

1. We deploy a DRL-based SAC policy on a real O-RAN E2 control loop and identify a structural difference from existing work [1, 2]. Only a per-slice resource ceiling is exposed, not a per-flow accept/reject primitive, so admission is realised as a ceiling adjustment.
2. We show that a QoE-aware reward significantly outperforms the SLA-only reward of [1, 2] and a static-at-cap arm on the same live O-RAN, with negligible control-loop overhead.
3. We evaluate a congested scenario offline and, at a smaller scale, live. This shows where the offline scarcity mechanism does and does not carry over to a deployed O-RAN.
4. This single-agent, single-gNB deployment validates the live control loop that the multi-agent GAT-MARL-FL framework in [3] would incorporate in future work.

II. Literature Review

This section reviews ten QoE-focused works from the verified pool of [3]. Recent DRL-driven slicing work has moved from SLA/QoS objectives toward explicit QoE. The authors in [5] used a hierarchical dueling DQN for user-centric bandwidth allocation, maximising satisfaction under stochastic requests while lowering latency. The researchers in [6] termed their formulation DRL-MQS, for multi-resource QoS satisfaction, reporting a 10.5% improvement in QoS-satisfaction ratio. Another study [7] trained a multi-agent Deep Deterministic Policy Gradient (DDPG) model for transmission-level QoE in wireless Virtual Reality (VR) over 5G New Radio (NR). The authors in [8] combined Convolutional Neural Network-Long Short-Term Memory (CNN-LSTM) prediction with Lyapunov-stable, non-DRL QoE-aware allocation. Another paper [9] instead used multi-agent Proximal Policy Optimisation (PPO), or MAPPO, for QoE across satellite-terrestrial Multi-access

M. P. Kunasegran, W. L. Pang, and S. K. Phang are with the School of Engineering, Taylor's University, Subang Jaya, Malaysia.
*Corresponding author.
ORCID: M. P. Kunasegran, 0009-0005-3169-1873; W. L. Pang, 0000-0001-8407-5648; S. K. Phang, 0000-0002-7877-8766.

Edge Computing (MEC) services. All five studies remain simulation-only.

A parallel thread infers QoE from network-side signals. The authors in [10] used a Transformer encoder-decoder to predict real-time-video QoE from low-level RAN information, cutting Mean Absolute Error (MAE) by 12.5–44%. Meanwhile, [11] embedded a gradient-boosting QoE estimator in a 5G O-RAN, reaching a coefficient of determination $R^2 = 0.986$. Both remain estimation only.

Closest to the present setting, three works place QoE control inside the O-RAN loop, none deploying a learned admission policy over a live E2 loop. The authors in [12] classified users by interference via an ML-based xApp, raising R-Factor scores in a simulated 5G HetNet. The researchers in [13] demonstrated near-real-time xApp RAN slicing, an open-loop Radio Resource Management (RRM) policy on an OAI plus FlexRIC testbed. FlexRIC is an open-source O-RAN near-real-time RIC. This work cuts overprovisioning by 30% and is the only one reviewed here validated on a live O-RAN, though unlearned. Meanwhile, [14] proposed a cross-layer QoE-driven bitrate heuristic for MEC video in simulation.

Across this literature, QoE is emerging as the primary objective and DRL the dominant method, yet none of the ten works reviewed here deploy a learned QoE-aware admission policy over a live E2 loop. Closing that gap is the central objective of this paper.

III. Methodology

This section describes the testbed and admission-control problem formulation, summarised by Fig. 1, with no new algorithmic framework proposed at this stage. Offline training against a closed-loop simulator, detailed in Section C, produces frozen checkpoints. These deploy on the live testbed below with no further learning, and any checkpoint failing offline convergence is discarded first. A held-out offline evaluation runs alongside this pipeline but does not rank-correlate with live outcomes.

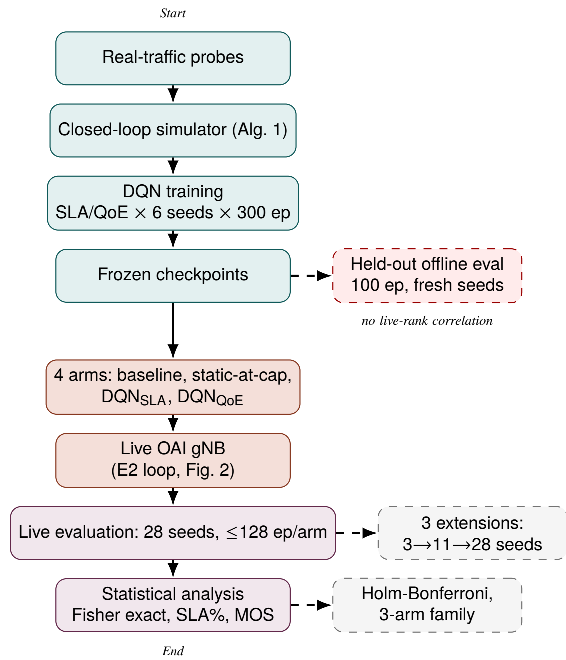

Fig. 1. End-to-end methodology, frozen checkpoints deployed live with no further learning, summarised in Table 1 and Figs. 4–5.

A. Testbed

Fig. 2 shows the Open5GS core connecting over N2/N3, carrying signalling to the Access and Mobility Management Function (AMF) and data to the User Plane Function (UPF). Both reach a real OAI gNB, an ORANSlice fork [15] with a simulated radio channel, rfsim. Its nominal capacity is 106 Physical Resource Blocks (PRBs) at $\mu = 1$, and its Medium Access Control (MAC) scheduler enforces the per-slice ceiling. Three `nr-uesoftmodem` UEs, one per slice, attach over the radio interface. The RIC-free E2 agent exchanges User Datagram Protocol (UDP) INDICATION/RESPONSE and CONTROL messages with the xApp/orchestrator, shared by all four arms. These messages carry `sst`, `sd`, `min_ratio` and `max_ratio`. Every step is logged to `omega_log.jsonl`.

Unlike the earlier simulated formulation [1, 2], the real O-RAN E2 interface exposes only a per-slice ratio ceiling on a 0–100 scale rather than a binary accept/reject action [4]. An accept decision raises a slice's ceiling toward a calibrated maximum, and a reject lowers it toward a floor. Ceiling values were calibrated by direct measurement, since the scale maps nonlinearly onto real PRBs here. The static baseline holds its ceiling fixed for the whole episode. *Static-at-cap* instead pins every ceiling at its calibrated maximum from the start, testing whether a generous fixed allocation alone matches a learned policy. Learned arms are DQN policies [16], all four issuing decisions through the same E2 agent/xApp loop.

B. Problem Formulation

The gNB state at time t is defined as

$$s_t = \left(\frac{U_s(t)}{B},\ C_t,\ \frac{L_s(t)}{L_{\max}}\ \forall s \in \mathcal{S} \right), \tag{1}$$

where $U_s(t)$ is the resource allocation of slice $s \in \mathcal{S} = \{\text{eMBB}, \text{URLLC}, \text{mMTC}\}$ and B is the total resource budget on the ceiling's 0–100 scale. C_t is the current congestion level, and $L_s(t)$ is slice s's queue length normalised by its capacity $L_{\max}$. The agent's action at each step is the ceiling adjustment described above. The SLA-only reward, used to train DQN-SLA, is

$$r(s_t, a_t) = \sum_{s \in \mathcal{S}} (\omega_s R_s - \lambda_s) - \mu C_t \|a_t\|_1, \tag{2}$$

where ω_s and λ_s are slice s's reward weight and SLA-violation penalty, and R_s is the reward for serving slice s. The term $\mu C_t \|a_t\|_1$ penalises total accepted volume by congestion level, matching existing work's SLA-driven reward structure [1, 2]. DQN-QoE instead trains against a differently-shaped reward,

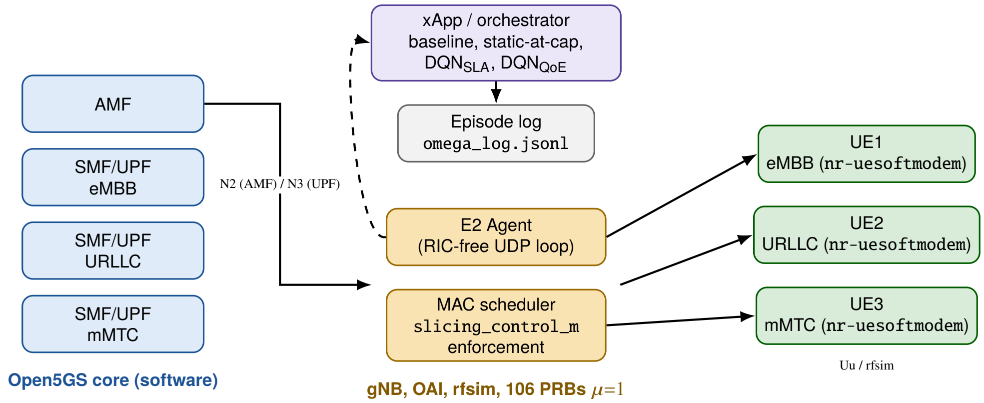

Fig. 2. Testbed software architecture.

our own prior work's eq. (9) [3], not an additive extension of Eq. (2).

$$r^{\mathrm{QoE}}(s_t, a_t) = \kappa\, \overline{\mathrm{MOS}}_t - \nu\, \mathrm{cost}_t - \xi\, \mathrm{viol}_t, \tag{3}$$

where $\overline{\mathrm{MOS}}_t \in [0,1]$ rescales the mean per-slice Mean Opinion Score (MOS) from Eqs. (4)–(5),

$$q_t = \max\left(\gamma\, \ell_t + \delta\, p_t + \epsilon/\tau_t,\ 0\right), \tag{4}$$

$$\mathrm{MOS}_t = \mathrm{clip}\big(\alpha - \beta \log(1 + q_t),\ 1,\ 5\big). \tag{5}$$

ℓ_t, p_t, τ_t are latency, packet loss and throughput. $\alpha, \beta, \gamma, \delta, \epsilon$ are coefficients calibrated in Section B. Above, cost_t reuses Eq. (2)'s congestion-scaled accepted volume $\mu C_t \|a_t\|_1$. The term viol_t is the mean per-slice violation severity, continuous unlike λ_s's flat charge. We compare both variants on the same live setup. We also report a priority-weighted compliance score $W = \sum_{s\in\mathcal{S}} \pi_s\, \mathrm{Compl}_s$, in place of an unweighted average. Here Compl_s is slice s's fully-compliant-episode rate, and $\pi_s = \omega_s$ is the same reward weight from Eq. (2), normalised so $\sum_s \pi_s = 1$.

C. Offline Training Environment

A DQN policy needs hundreds of episodes per seed to converge, 1,800 across this study's six checkpoints alone, far more live rig time than this paper's entire live evaluation uses. Training therefore happens offline, against a synthetic, closed-loop source. Served traffic is capped by the slice's most recent ceiling, and unserved demand persists as backlog. An admission decision therefore carries a real, delayed consequence, unlike an earlier open-loop source where always accepting was trivially reward-optimal. Algorithm 1 summarises the closed-loop update, one polling step per gNB/slice pair.

This source's demand mean is tied to offered-load levels measured on this rig, but its temporal dynamics are not validated against live traffic. Held-out evaluation of the same six checkpoints does not rank-correlate with live compliance. This traces in part to the simulator's backlog-capacity parameter, which was never validated against real SLA-margin magnitude. Offline training still produces competent policies but cannot rank or pre-select which checkpoint will succeed live, so every number in this paper's Results is a live measurement, not an offline projection.

Algorithm 1 Offline closed-loop 5G traffic simulation (one polling step, all gNBs/slices)

Require: per-(gNB,slice) state: offered, backlog, ceiling, UE set
for all gNB g, slice s **do**
  $mean \leftarrow$ mean_offered_ratio$_s$ $\times B \times$ gnb_load$_g$
  offered$_{g,s} \leftarrow \max(0,\ \text{offered}_{g,s} + 0.1(mean - \text{offered}_{g,s}) + \text{noise})$
  if a reject freed backlog on the previous step **then**
    backlog$_{g,s} \leftarrow \max(0,\ \text{backlog}_{g,s} - \text{relief})$
  end if
  $demand \leftarrow$ offered$_{g,s}$ + backlog$_{g,s}$
  $served \leftarrow \min(demand,\ \text{ceiling}_{g,s})$
  backlog$_{g,s} \leftarrow \min(\text{backlog_cap},\ demand - served)$
  with prob. churn_prob, replace one UE's RNTI in (g, s)
  $bler \leftarrow 0.02 + 0.3 \times (\text{backlog}_{g,s}/\text{backlog\_cap})$
  for all UE u in (g, s)'s set, $n \leftarrow |\text{UEs}_{g,s}|$ **do**
    emit KPM sample for u: $served/n$, $bler$, backlog$_{g,s}/n$
  end for
end for
agent observes the resulting state s_t
agent sends ceiling$_{g,s}$ $\forall (g, s)$ for the next step

D. Evaluation Setup

Each of the four arms was evaluated live across 28 random seeds, with frozen policy weights, reached in three extensions. The first added 3 seeds at 2 episodes each. The second added 8 more at the full 5-episode protocol, once a QoE-calibration issue detailed in Section B was corrected, for a running total of 46 episodes/arm. The third added 17 more, 16 at 5 episodes and one at 2, for the fully powered 128 episodes/arm reported here. This sample is large enough that a tied Fisher exact test is genuine, not underpowered. The static-at-cap arm reuses the static baseline's control code with only the ceiling changed. The DQN arms followed the offline training procedure, 300 episodes per seed.

IV. Results and Discussion

A. Live Testbed Results

We classify a live episode as *fully compliant* if every per-step SLA margin stays positive for all three slices, and *collapsed* otherwise. Each margin is sampled every 5 seconds and taken as whichever of queue-occupancy or packet-loss is worse. Table 1 summarises SLA compliance for all four arms across 128 live episodes each, under the corrected QoE calibration from Section B. DQN-QoE sustains 100% compliance on every slice, every episode, 128 of 128. DQN-SLA and static-at-cap reach an identical, slightly lower 126/128. The static baseline trails both at 120/128, its ceiling staying flat rather than rising toward the calibrated maximum.

Fig. 3 plots DQN-QoE against DQN-SLA on the one seed where DQN-SLA collapses. Between steps 33–36, it drops URLLC's ceiling as low as half its cap before recovering, while DQN-QoE holds cap throughout. This is the mechanism behind that episode's 0% vs. 100% URLLC outcome, raising whether DQN's advantage reduces to a more generous fixed ceiling. An earlier, smaller-sample evaluation suggested static-at-cap collapses more often than DQN-SLA. The gap was 27% vs. 0%, $p = 0.0149$. At this paper's 3× larger sample, that edge does not hold. Both reach an identical 126/128, $p = 1.0$, a genuine null result. Both separate further from the baseline's 120/128 than at 46 episodes/arm, but only DQN-QoE's gap is significant. Raw $p = 0.10$ for DQN-SLA and static-at-cap vs. baseline, identical at 126/128 each, and $p = 0.0070$ for DQN-QoE vs. baseline become $p = 0.205$ and $p = 0.021$ after correction. The QoE-aware reward's advantage survives correction, but the SLA-only reward's does not.

Fig. 4 shows this episode-by-episode, one point per episode, DQN-QoE at 100% throughout and the other three in the same bimodal split. Fig. 5 shows two bars per arm, only DQN-QoE's coinciding at 100%. We also trained 5 more DQN-SLA checkpoints at a reduced 21-episode protocol. Despite indistinguishable offline convergence, live records varied substantially. Of the five, 3 were perfect and one reached 19/21. The remaining checkpoint fell to 13/21, worse than baseline. This observation confirms that live robustness is not predictable from offline convergence alone, as Section C details.

B. *Deployment Feasibility*

Measured against the real gNB, the E2 round trip has a median latency of 0.57 ms. Its 90th/99th percentiles, the worst 10%/1% of calls, reach 1.10 ms and 1.77 ms. The DQN model, 18,562 parameters and ~300 KB on disk, selects an action with a median CPU inference latency of 67.9–68.3 μs. p90/p99 reach 84.6–85.1/108.9–115.0 μs across both reward modes, a small fraction of the 5-second control interval. The step's actual dominant cost is the QoE-mapper Long Short-Term Memory (LSTM) inference at ~7.3 ms, an order of magnitude above the DQN decision at ~0.07 ms or the E2 exchange at ~0.6 ms, all three staying under 2% of the step budget. The configured 5.0 s control cadence lands at 5000.7 ms, a 0.7 ms gap. Comparing offline predictions against live results showed a good match for eMBB but not URLLC/mMTC, trained under a different demand assumption than observed live, and surfaced a QoE-calibration bug. The MOS estimator's throughput term used a PRB-usage ratio at run time, but eMBB's coefficients assumed an absolute per-UE PRB count. This meant 0.15 read as almost-starved, pinning eMBB's MOS at the scale floor. Refitting those coefficients against the ratio actually used lifted eMBB's MOS from a uniform 1.00 to 2.29, 3.62, and 4.11. This predates Table 1, unaffected by the MOS estimate.

TABLE 1. Live SLA compliance (%), 128 episodes/arm

Arm	eMBB	URLLC	mMTC
Static	94.0	93.8	93.9
Static-at-cap	99.1	98.4	98.5
DQN (SLA reward)	98.8	98.4	98.5
DQN (QoE reward)	**100.0**	**100.0**	**100.0**

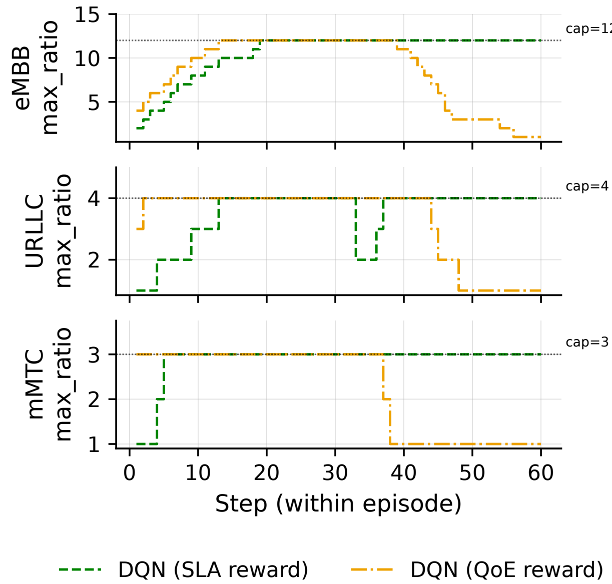

Fig. 3. Commanded ceiling per slice, DQN-SLA vs. DQN-QoE, seed 955, episode 1.

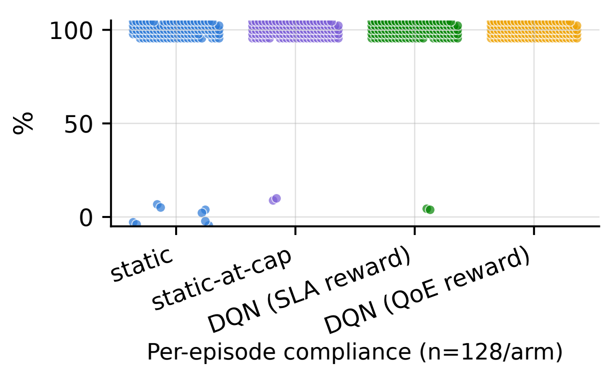

Fig. 4. Per-episode SLA compliance (%), one point per episode.

C. *Congested Scenario, Offline Results and a Live Pilot*

Unlike the constant demand of Sections A and B, we evaluate the same checkpoints against a congested extension of Section C's offline source, where demand exceeding the shared pool scales down each slice's allocation proportionally. Pooled over 165 held-out episodes/arm across 11 seeds, Table 2 shows both DQN variants raising URLLC compliance above the baseline. Both trade off eMBB and mMTC compliance, matching existing work's URLLC-first behaviour [1, 2]. Applying W,

TABLE 2. Congested offline compliance (%), pooled 165 ep/arm

Arm	URLLC	eMBB	mMTC	*W*
Static baseline	19.7	32.7	19.8	**24.9**
DQN (SLA reward)	27.7	7.7	9.4	19.1
DQN (QoE reward)	25.0	8.2	11.6	17.9

with $\omega_{\mathrm{URLLC}} = 5.0$ and $\omega_{\mathrm{eMBB}} = 3.5$, the baseline still scores higher overall. URLLC carries the larger weight, but the DQN arms gain only 5–8 points of URLLC compliance against a 24–25 point eMBB loss, so eMBB's larger raw swing outweighs URLLC's smaller weighted gain. We also live-evaluated the same checkpoints at a smaller pilot scale, one seed and two episodes per arm. Every arm was fully SLA-compliant live, a scale mismatch since this rig's ceilings never approach the offline budget's contention. The QoE-aware reward still produced a materially higher URLLC MOS of 4.83 out of 5 than the SLA-only reward's 2.44 or the static baseline's 2.66, at the same 100% compliance.

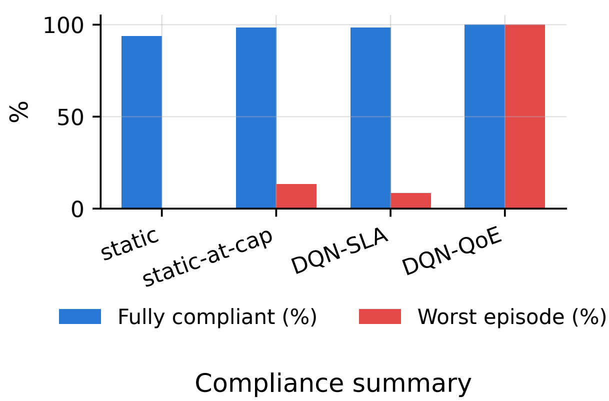

Fig. 5. Compliance summary, fraction of episodes fully SLA-compliant and worst single-episode compliance, by arm.

V. Conclusion and Future Work

This paper moved an existing simulated DRL-based SAC formulation onto a real OAI gNB over its native E2 control loop, with the radio channel still simulated. A fully powered live evaluation across 128 episodes per arm confirmed the QoE-aware reward's significant compliance advantage over the static baseline (Table 1), with the SLA-only reward and static-at-cap policy only numerically ahead of it, and negligible control-loop overhead throughout. Under offline congestion, the learned policy prioritises URLLC unless priorities are weighted. No live arm faced that tradeoff, though the QoE-aware reward gave URLLC a better experience score.

Our future work will improve the offline environment's fidelity to real traffic and extend this deployment toward the fully-fledged GAT-MARL-FL framework from our earlier review. This leaves topology-scalability, federated-learning privacy and disruption-resilience open.

Acknowledgment

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